

time nearer than you do. Also I have made the same inquiry of some comets who pass my way. All have answered as I have done. I believe that even the sun himself and every star would answer in the same way.

EARTH. Despite all that you say I have hopes. Today in particular men promise me much happiness in the future.

MOON. Hope as much as you like. And I promise you that you can hope forever.

EARTH. You know what? These men and these animals are starting to make a noise, for at the spot from which I'm con-

versing with you it's night, as you see, or rather as you don't see. They were all sleeping and at the din which we are making with our talking they are awakening in great fear.

MOON. But here, as you see, it's day.

EARTH. Now I don't want to be the cause of terrifying my folk and of interrupting their sleep, which is the greatest good they have. However, we'll talk to each other again some later time. So good-bye. Good day.

MOON. Good-bye. Good night.

GIACOMO LEOPARDI: DIALOGUE BETWEEN FREDERICK RUYSCH AND HIS MUMMIES¹

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CHORUS OF CORPSES IN FREDERICK RUYSCH'S LABORATORY²:

*"In you, alone eternal in the world, to whom
Turns every created thing,
Rests, O Death,
Our shorn being,
Not happy, but safe
From old griefs. Deep night
Dims the sober thought
Of our confused mind;
Our barren spirit feels itself
Too weak to hope, to wish.
Thus, free of sorrow and of fear,*

*It consumes the slow and empty
Ages without ennui.
We lived; and as the confused memory
Of a hideous ghost
Or of a feverish nightmare
Wanders in the suckling infant's soul,
So lingers the memory
Of our life; but the memory
Is remote from fear. What were we?
What was that bitter point
Which had the name of life?
A secret and stupefying thing
Is life now to our thoughts, and such as
Unknown death seems
To the thought of the living. As from death
Our live being flees, so flees
From the vital flame
Our shorn being,
Not happy, but safe.
However, to be blessed
Is denied by Fate to mortals and to the
dead."*

¹ This translation of Leopardi's *Dialogo di Federico Ruysch e delle sue mummie* is by Rufus Suter (deceased, February 29, 1968). The dialogue appeared originally in 1827. The text used for this translation is in Leopardi's *Opere morali*, Rizzoli-Editore, Milan (Biblioteca Universale Rizzoli, 327-329), first ed., 1951, pp. 130-136. The following notes, unless otherwise indicated, are taken from the text.

² "In regard to these mummies, which in scientific language would be called anatomical preparations, see Fontenelle, *Éloge de Mons. Ruysch*" (Leopardi). Ruysch was a physician of Amsterdam (1638-1731), who succeeded in preserving cadavers from corruption.

RUYSCH (OUTSIDE HIS LABORATORY
LOOKING THROUGH THE CRACKS OF THE
DOOR). The deuce! Who has taught music

to these corpses which sing at midnight like roosters? Really, I'm in a cold sweat. I'm almost more dead than they. I didn't think that because I preserved them from decay they would do me the trick of coming back to life. But so it is. With all my philosophy I'm shivering from head to foot. A curse on the demon who tempted me to put this tribe in my house! I don't know what to do. If I leave them shut up here perhaps they will break the door down, or come out through the key-hole, and pay me a visit in bed. To cry for help on account of fear of corpses isn't appropriate in my case. Well, let's be brave and let's try to frighten them.

(ENTERING). Boys, what game are we playing? Don't you recall that you are dead? What is this hubbub? Have you perhaps become swelled-headed because of the Czar's visit,³ and think that you aren't subject to the same laws as before? I suppose you meant to play a prank and that you aren't serious. If you have come back to life, congratulations! But I don't have the money to pay the expenses of the living as I did of the dead. So away with you! If what is said about vampires is true, and you are of them, look for other blood to drink. I'm not inclined to let mine be sucked as generously as I put that sham blood in your veins. In a word, if you are willing to stay as quiet as you were before, well and good. But otherwise I warn you that I'll take the door-bar and kill you all.

CORPSE. Do not be angry. I promise you that we all shall stay as dead as we are without your killing us.

RUYSCH. Then why that whim of singing?

CORPSE. A short time ago, precisely at midnight, that Great Mathematical Year⁴

³ "Ruysch's laboratory was visited twice by Czar Peter I, who bought it and had it transported to Petersburg" (Leopardi). [*Tr.'s note*: Perhaps Ruysch's method was that subsequently used by the Russians to preserve the bodies of recent political leaders].

⁴ The ancients called the year in which the stars, having completed their revolution, were again all at the same point from which they had

of which the ancients wrote so much was completed for the first time. And also this is the first time the dead have spoken. Not only we, but in every cemetery, in every sepulcher, at the bottom of the sea, under the snow and the sand, beneath the open sky, and everywhere that the dead are, they, like us, have sung the same little song which you heard, at midnight.

RUYSCH. How long will they continue to sing and talk?

CORPSE. They have already stopped singing. They have the power to speak for a quarter of an hour. Then they will become silent again, until the Great Year fulfills another circuit.

RUYSCH. If such be the case, I believe that I shan't be disturbed a second time from my sleep. Just keep on talking freely among yourselves. I'll gladly stand by and listen, out of curiosity; and I shan't disturb you.

CORPSE. We cannot speak except to answer a live person. Those of us who do not have the opportunity to answer a live person remain silent after we have finished our song.

RUYSCH. What a pity! I imagine it would have been fun to listen to you talking to each other, if you had been able to do so.

CORPSE. Even if we had been able to do so, you would have heard nothing, because we should have had nothing to say.

RUYSCH. A thousand questions occur to me to ask you. But the time is too short to allow choice. So, in a word, tell me what feelings you had in your mind and body at the moment when you died.

CORPSE. I was not aware of the precise moment when I died.

OTHER CORPSES. Nor were we.

RUYSCH. Why were you unaware?

CORPSE. For the same reason that you are unaware of the moment when you fall asleep, for instance, regardless of how attentive you are.

started, the Great Year, or the Mathematical Year (see Cicero, *De Natura deorum*, II, ch. 20).

RUYSCH. But falling asleep is natural.

CORPSE. Does not dying seem natural to you? Show me a man, or an animal, or a plant that does not die.

RUYSCH. Since you weren't aware of dying, I'm not surprised that you keep on singing and talking. "Thus he, unaware of the blow, kept on fighting and he was dead,"⁵ says an Italian poet. On this business of dying I used to think that your peers knew more than the living. But to get down to facts: Didn't you feel some pain when you died?

CORPSE. What pain can it be of which a person is unaware when he feels it?

RUYSCH. But everybody is convinced that dying feels exceedingly painful.

CORPSE. As if death were a feeling, and not the contrary of feeling.

RUYSCH. Those who have the same notions about the soul's nature as the Epicureans,⁶ and those who hold the common opinion⁷—all, or almost all, agree with me: death is by its very nature incomparably painful.

CORPSE. Well then, will you please ask these questions for our sake? If a man does not have the power to know the moment when mere sleep, lethargy, syncope, or some other cause temporarily and partially interrupts his vital operations, how can he know the moment when the same operations cease entirely and forever? How can a live feeling have a place in death? How can death by its own quality be a live feeling? When the power or feeling is not only weakened and scarce but is reduced to so tiny a thing that it is wanting and canceled altogether, how can a person have a strong feeling? How can the extinction of the power of feeling be an extremely vivid feeling? Moreover, anybody can see that those who are dying of an acute and pain-

ful disease become tranquil and rest as death approaches, shortly before they expire. Their severely reduced life is insufficient for pain. The pain ceases before the life. Will you please speak thus, for our sake, to all those who believe that they must suffer when they die?

RUYSCH. Your reasoning might convince an Epicurean—but it won't convince those who hold the common opinion of the soul's substance, as I did in the past, and as I shall do even more from now on, since I've heard the dead talk and sing. We who hold the common opinion believe that dying consists in the separation of soul and body. And we don't understand how these two entities, joined together, glued together as it were, so that they make up a single person, can be separated without extreme violence and suffering.

CORPSE. Tell me: Is the spirit hung on the body by some nerve or muscle or membrane, which has to be broken when the spirit departs? Or is it a limb of the body which must be forcibly split off, or cut away? Do you not see that the soul goes out of the body only when it is prevented from staying in it and no longer has a place there, and not at all because of any force that jerks it away or uproots it? Tell me: When the spirit entered your body did you feel it driving itself in, or powerfully linking itself with your body, or, as you say, becoming glued to it? Why then should you feel a violent sensation when it detaches itself from your body and goes out? Rest assured that the soul's entrance and exit are equally tranquil, facile, and gentle.

RUYSCH. Then if death isn't pain, what is it?

CORPSE. A pleasure more than anything else. Know that dying, like falling asleep, doesn't happen at a single instant, but by steps. It is true that these steps are more or fewer, and larger or smaller, according to the cause and kind of death. At the final instant death brings neither pain nor pleasure, as sleep does not. In the preced-

⁵ Francesco Berni in the adaptation of *Orlando furioso*, canto 53, stanza 60.

⁶ The Epicureans, or followers of Epicurus (342–270 B.C.), since they believed everything to be composed of matter, including even the soul, held that the soul dies with the body.

⁷ The common opinion is that of those who believe in the soul's immortality.

ing instants it cannot generate pain because pain is a live thing, and a man's senses at the moment when death has begun are moribund, which is to say that their powers are extremely attenuated. The approach of death can be the occasion of pleasure because pleasure is not always a live thing. Actually, perhaps most human enjoyment consists in some sort of torpor. Thus a man's senses can feel pleasure even when they are on the verge of extinction. Very frequently torpor itself is a pleasure, especially when it frees from suffering. As you well know, the cessation of pain or discomfort is in itself a joy. Hence the torpor of death is bound to be welcome in proportion as it frees a man from greater suffering. As for me, although at the hour of death I did not pay much attention to what I was feeling, because the physician had forbidden me to use my brain, still I remember that my feeling was not much different from the enjoyment which the torpor of sleep brings one who is falling asleep.

OTHER CORPSES. Also we seem to recall the same.

RUYSCH. Let it be as you say, though

everybody with whom I have had the opportunity to discuss this matter judged differently. But, as far as I can remember, nobody alleged personal experience. Now tell me: At the time when you died, while you were feeling that sweet sensation, did you believe that you were dying, and that your pleasure was a courtesy of death? Or did you imagine something else?

CORPSE. Until I was dead I was never convinced that I should not escape that danger. Right down until the last moment of my power to think I hoped that at least an hour or two would be spared me, as I judge happens to many while they are dying.

OTHER CORPSES. The same happened to us.

RUYSCH. Thus Cicero says that nobody is so decrepit that he doesn't promise himself to live at least a year. Well, how did you at last know that you were dead? They do not answer. Boys, don't you understand me? The quarter of an hour must have passed. Let's touch them. They really are dead again. There's no chance that they'll frighten me a second time! Let's go to bed.

MODELS AND THE CURRICULUM

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INTRODUCTION

RECENTLY I had the opportunity to overhear some comments by elementary teachers who were reacting to a lecture on Inquiry Training delivered by a science educator at a local county-level conference. They were discussing what method each utilized in the teaching of science. The responses ranged from the Hickory Dickory Dock-Science by the Clock Method, the Inquiry-Training Method, the Problem-Solving Method, to the Inquiry-Discovery Method.

While the members entering into this discussion did not stop to inquire as to the assets, liabilities, uniquenesses, or the relationships of one approach to the other, they later chorused, "What did he (the lecturer) mean when he talked about "model building?" "Is this another safari into the jungle of scientific folderol?"

This paper is an attempt to partially cover the topic of models.

Models In The Curriculum

Recognition of the merits of model building is borne out by the integration of